Zehan Li

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Personal Website | Google Scholar

EDUCATIONAL BACKGROUND

- Sep/2025 - Current. **University of Chicago** — Chicago, IL, USA
  M.A. in Computational Social Science — *GPA: 3.73/4.00*
- Jan/2021 - Jun/2025. **University of California, San Diego** — San Diego, CA, USA
  B.S. in Cognitive Science (Machine Learning Track) with Highest Distinction — *GPA: 3.61/4.00*
  B.S. in Applied Mathematics
  Minor in History; Minor in Linguistics

Related Courses: Neural Networks and Deep Learning (A+); Computational Linguistics (A+); Machine Learning with Few Labels (A); Advanced Machine Learning Methods (A+); Modeling & Data Analysis (A).
Honors: **Provost Honors** (6 terms), **Triton Research & Experiential Learning Scholars** (2 terms)

RESEARCH INTERESTS

I study what language models do to human meaning and judgment as they increasingly sit between people: what they compress, add, or distort as they take part in creative expression and cultural transmission, and how they reshape attention and evaluation when people and institutions rely on them. I aim to build human-centered AI and evaluations that capture what traditional benchmarks miss.

PUBLICATIONS

- **Zehan Li**, Yutong Zhu, Siyang Wu, Honglin Bao, James A. Evans. Narrative Flattening: How Post-Training Compresses Thematic, Affective, and Stylistic Variation in LLM Fiction. **EMNLP 2026**, Main Conference, **Oral**.
- **Zehan Li**, Yuxuan Wang, Ali El Lahib, Ying-Jieh Xia, Xinyu Pi. Simulated Ignorance Fails: A Systematic Study of LLM Behaviors on Forecasting Problems Before Model Knowledge Cutoff. **IJCAI 2026**, Main Track.
- Ali El Lahib, Ying-Jieh Xia, **Zehan Li**, Yuxuan Wang, Xinyu Pi. Temporal Leakage in Search-Engine Date-Filtered Web Retrieval: A Retrospective Forecasting Case Study. **ACL 2026**, Main Conference.
- **Zehan Li**, Ruhua Pan, Xinyu Pi. Beyond LLMs: A Linguistic Approach to Causal Graph Generation from Narrative Texts. **WNU @ NAACL 2025** (Workshop).

RESEARCH EXPERIENCES

- What happens to meaning when LLMs translate across languages and cultures? — Jun/2026 - Current
  *Project Lead. With **Junsol Kim** & **Prof. James A. Evans**, University of Chicago*
  - **Built a per-dimension semantic atlas of cross-lingual transmission** across lexical, grammatical, and pragmatic dimensions (e.g., kinship, evidentiality, honorifics) and six model families.
  - Showed that translation compresses meaning in one direction and forces new commitments in the other, and that surface-form loss is not semantic loss; now tracing how meaning drifts as texts are relayed across multiple languages.
- What happens when AI sits on both sides of a hiring decision? — Jun/2026 - Current
  *Project Lead. With **Prof. Jiaxin Pei** (UT Austin) & **Prof. Xinyu Liang** (INSEAD)*
  - **Built a three-level audit of LLM résumé screening**, from a single résumé to a multi-candidate prompt to an autonomous agent, on the same real résumés and multiple frontier LLMs, placing every design choice on a common rerun-to-chance scale.
  - Found that screening decisions are highly unstable and shaped by many factors: reversing the display order of 100 résumés in one prompt replaces 74% of the top-3 shortlist (vs. 26% on rerun), and 30% of résumés pass under some LLMs but fail under others.

- How do LLMs remember, and reshape, the scientific literature? Oct/2025 - Current
  *Researcher. Supervised by **Prof. Bernard Koch**, University of Chicago*
  - **Found that LLM citation memory is organized by textual co-occurrence** (semantic similarity, co-citation) rather than citation-graph topology or the social processes that generate citations.
  - Designed a citation in-painting task across more than 10 LLMs on ∼100K SciSciNet v2 papers, using Infini-gram corpus frequencies to separate model behavior from data composition.
  - Now comparing LLM-generated bibliographies against real human citations for the same papers.
- Why do LLM-generated stories feel narratively flat? Jan/2026 - May/2026
  *Project Lead. Supervised by **Prof. James A. Evans**, Knowledge Lab, University of Chicago*
  - Resulting publication: EMNLP 2026 (Oral). **Led a matched story-continuation study of how post-training reshapes LLM fiction**, comparing Base, SFT, DPO, and RLVR checkpoints against human continuations across multiple story corpora.
  - Developed computational measures of thematic motion, affective intensity, and stylistic diversity, showing that post-training progressively compresses narrative variation into more uniform and neutral outputs.
- Can prompts reliably rewind LLM knowledge for forecasting evaluation? Jul/2025 - Jan/2026
  *Project Lead. Supervised by **Prof. Zhiting Hu**, UCSD*
  - Resulting publications: IJCAI 2026 and ACL 2026. **Provided the first systematic test of whether Simulated Ignorance (SI) can approximate True Ignorance (TI) in retrospective LLM forecasting evaluation**, deploying `Qwen3-32B` and `Llama` models on cluster servers.
  - Across 477 competition-level questions and 9 models, found that SI fails systematically: cutoff instructions leave a 52% performance gap, chain-of-thought cannot suppress post-cutoff knowledge, and reasoning-optimized models are *worse*, showing that prompts cannot reliably "rewind" model knowledge.
- Quantifying Conceptual Density and Vacuity in Text Aug/2024 - Jun/2025
  *Undergrad Honors Thesis. Supervised by **Prof. Zhuowen Tu** & **Alex Warstadt**, UCSD*
  - **Developed a theoretical model for quantifying text specificity and vacuity** using linguistic properties as primary dimensions, proposing a metric system for conceptual density in written communication.
  - Benchmarked outputs from multiple LLMs to compare their conceptual density and informational richness against human-written text.
- Can linguistic properties improve causal graph generation from narratives? Feb/2024 - Sep/2024
  *Project Lead. Supervised by **Prof. Zhiting Hu**, UCSD*
  - Resulting publication: WNU @ NAACL 2025. **Developed an open-source linguistics-based model for causal graph generation**, achieving 82% interpretability, a 9% improvement over direct GPT prompting baselines.
  - Built a feature extraction pipeline using `spaCy` and `nltk` that leverages linguistic properties (e.g., time boundness) for causal link identification, yielding a 15% accuracy gain over LLM-only approaches based on human evaluation.

EXTRACURRICULAR EXPERIENCES

- Metaculus AI Forecasting Tournament (Fall 2025 Cycle) Oct/2025 - Dec/2025
  - Developed an autonomous AI forecasting agent that competed in the Metaculus AI Bot Tournament, achieving **15th place globally** and a $1,500 prize.
  - Engineered a well-calibrated probabilistic prediction system, demonstrated by close alignment between predicted confidence and actual resolution rates across the full probability range.

PROFESSIONAL EXPERIENCES

- Instructional Assistant (2 courses): Deep Learning & Supervised Machine Learning Sep/2024 - Jun/2025
  *UCSD Cognitive Science Department. Under **Prof. Zhuowen Tu***
  - Led weekly 80-minute discussion sessions for 100+ students across two quarters, covering topics from regression and classification to GPT architectures and prompt engineering.
  - Provided individualized support through office hours and detailed assignment feedback.

OTHER INFORMATION

- Computer Skills: Python, Java, SQL, R, MATLAB, JavaScript, LaTeX, HTML/CSS, Docker
- Languages: Chinese (native), English (proficient), French(Intermediate)